

12.100

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Question:

A Markov chain with state space $(0, 1, 2)$ has transition matrix

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \quad (1)$$

Which Statements are true?

- 1) 0 and 1 are recurrent
- 2) 2 is transient
- 3) Chain has unique stationary distribution
- 4) Chain is irreducible

Solution:

Markov Chain:

The probability of going next state j only depends upon current state i .

$$P(X_{n+1} = j | X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j | X_n = i) \quad (2)$$

Transition Matrix:

$$\mathbf{P} = [p_{ij}] \quad (3)$$

$$p_{ij} = P(X_{n+1} = j | X_n = i) \quad (4)$$

$$\sum p_{ij} = 1 \quad (5)$$

Recurrent: Recurrent state is the one that the chain will return to with probability 1.

Transient: There is non zero probability that it will never return.

As

$$p_{01} > 0, p_{10} > 0 \quad (6)$$

0 and 1 are recurrent as they are closed communicating class. As

$$p_{02} = 0, p_{12} = 0 \quad (7)$$

2 is transient

Stationary distribution: Stationary distribution is π such that

$$\pi \cdot \mathbf{P} = \pi, \sum \pi_i = 1 \quad (8)$$

$$\implies \pi \cdot (\mathbf{P} - \mathbf{I}) = 0 \quad (9)$$

$$(\mathbf{P}^T - \mathbf{I}) \cdot \pi^T = 0 \quad (10)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ \frac{-1}{2} & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{-1}{2} & \frac{2}{3} \end{pmatrix} \cdot \pi^T = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (11)$$

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{2}{3} & 0 \end{array} \right) \quad (12)$$

$$R_2 \longrightarrow R_2 + R_3/2$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{3} & 0 \\ 0 & 0 & -\frac{2}{3} & 0 \end{array} \right) \quad (13)$$

$$\pi^T = \left(\frac{1}{2} \quad \frac{1}{2} \quad 0 \right) \quad (14)$$

Therefore Chain has unique stationary distribution .

Irreducible chain: If for every j there exist $(\mathbf{P}^n)_{ij} > 0$

the chain is not irreducible as 2 for j=2 ,The condition is not satisfied